



FUTURE INTERNS

CYBER SECURITY

TASK2_REPORT

SOC&SIEMLogAnalysisUsing Splunk

Submitted by:

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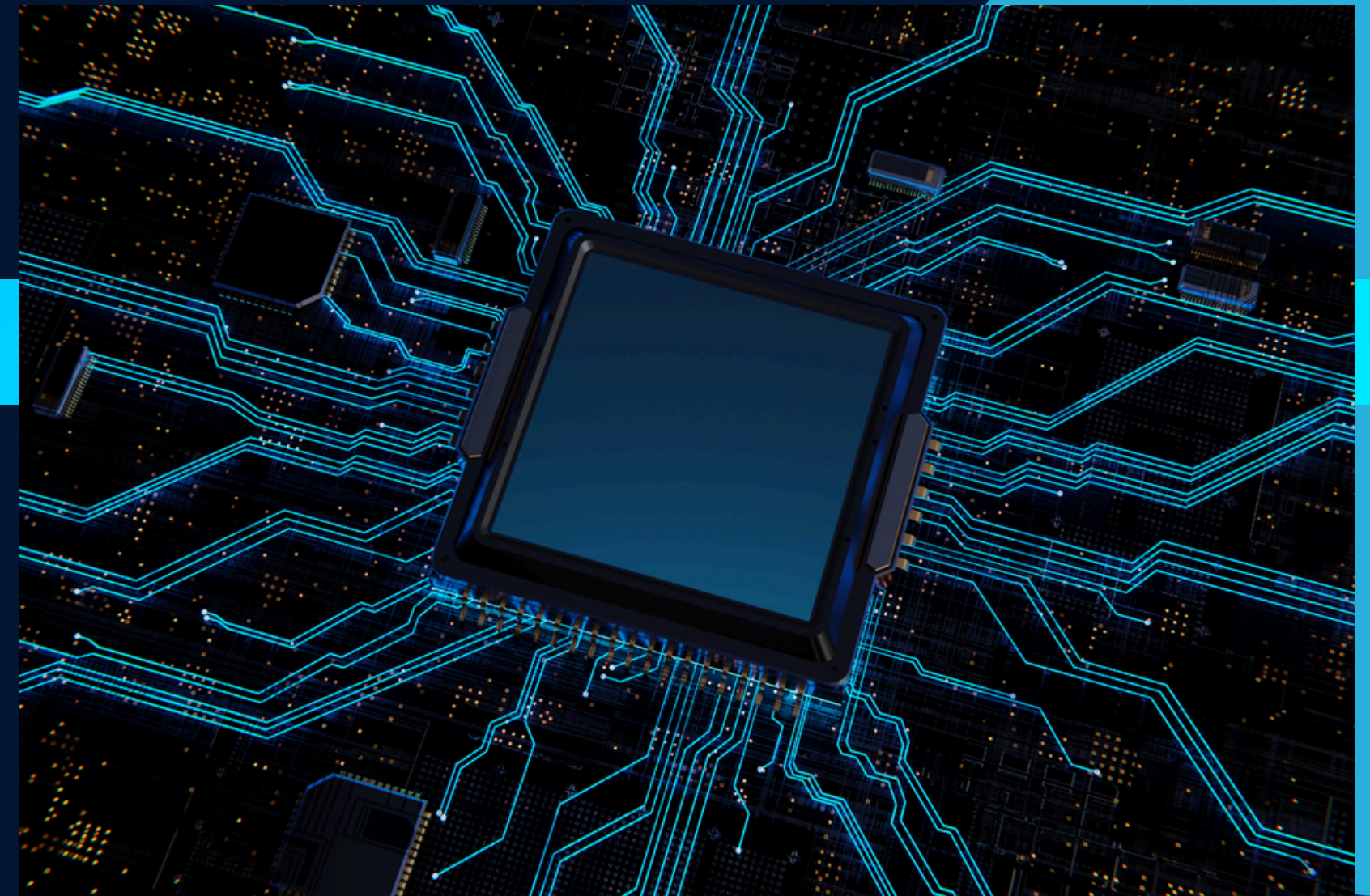
FUTURE_CS_02

INTRODUCTION

- This project focuses on analyzing security logs using Splunk SIEM platform.
- The objective is to identify suspicious activities, monitor security alerts, and understand basic incident response procedures.

OBJECTIVE

- Analyze security logs
- Detect suspicious activities
- Identify security alerts
- Perform basic incident response analysis

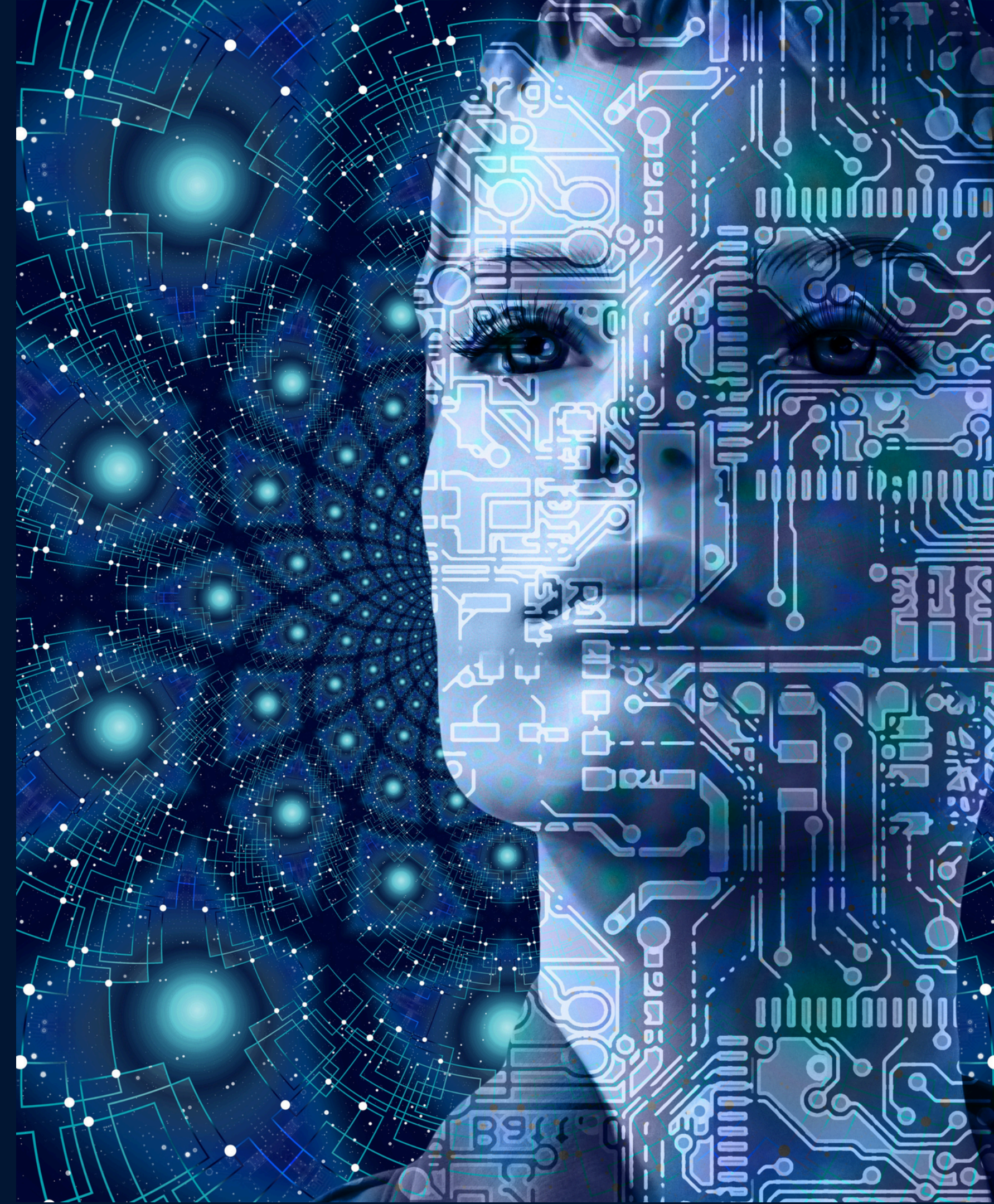


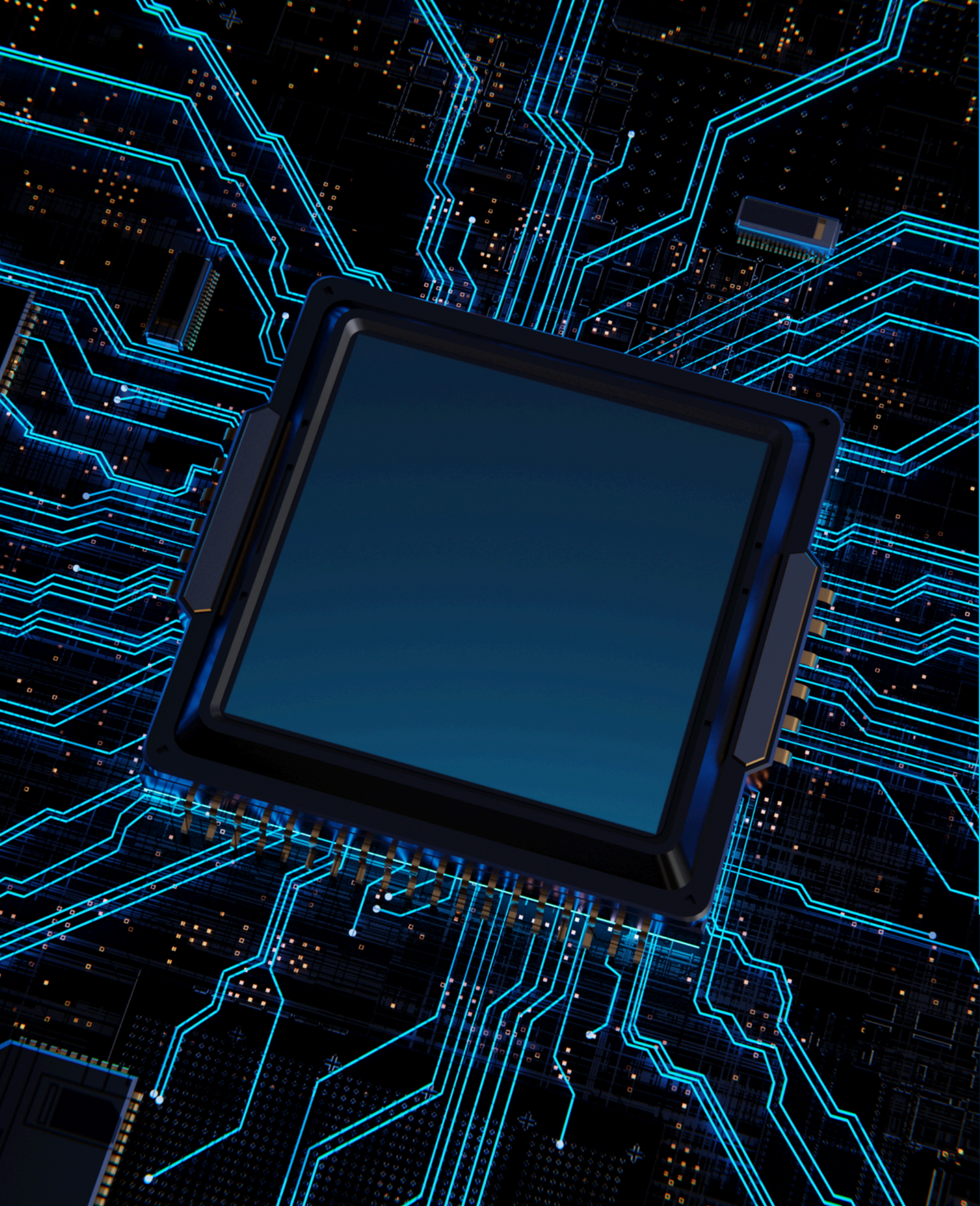
Tools Used

Tool	Purpose
Splunk Enterprise	Log Analysis & SIEM Monitoring
Sample Logs	Security Event Simulation

Log Analysis

The uploaded logs were analyzed using Splunk Enterprise. Multiple suspicious activities such as failed login attempts, malware detection, and unauthorized access attempts were identified during analysis.





DETECTED ALERTS

Alert	Severity	Status
Failed Login Attempts	Medium	Detected
Malware Detection	High	Detected
Unauthorized Access Attempt	High	Detected

Search | Splunk 10.4.0

localhost:8000/en-US/app/search/search?q=search%20source%3D%20Sample_logs.txt%20host%3D%20Jaya_Kondreddy%20sourcetype%3D%20Sample_logs&earliest=0&latest=&sid=17797051...

splunk>Enterprise

New Search

source="Sample_logs.txt" host="Jaya_Kondreddy" sourcetype="Sample_logs"

Time range: All time

✓ 10 events (before 5/25/26 4:02:47.000 PM)

No Event Sampling

Job

Events (10)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

Zoom to Selection

Deselect

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 10

a date_month 1

date_second 10

a date_wday 1

date_year 1

a date_zone 1

i	Time	Event
>	5/25/26 9:19:22.000 AM	May 25 09:19:22 Security alert generated for abnormal system activity host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:18:40.000 AM	May 25 09:18:40 User account locked after repeated failed logins host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:17:09.000 AM	May 25 09:17:09 Possible brute-force attack detected host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:16:31.000 AM	May 25 09:16:31 Firewall blocked suspicious traffic activity host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:15:52.000 AM	May 25 09:15:52 Multiple failed login attempts from 203.0.113.45 host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:10:15.000 AM	May 25 09:10:15 Failed login attempt from 192.168.1.12 host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs

 RESULTS

Search | Splunk 10.4.0

localhost:8000/en-US/app/search/search?q=search%20source%3D%20Sample_logs.txt%20host%3D%20Jaya_Kondreddy%20failed%20login&earliest=0&latest=&display.page.search.mode=smart&di...

splunk>Enterprise

New Search

source="Sample_logs.txt" host="Jaya_Kondreddy" failed login

Time range: All time

✓ 2 events (before 5/25/26 4:13:49.000 PM)

No Event Sampling

Job

Events (2)

Patterns

Statistics

Visualization

Timeline format

Zoom Out

Zoom to Selection

Deselect

Format

Show: 20 Per Page

View: List

< Hide Fields

All Fields

SELECTED FIELDS

a host 1

a source 1

a sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 2

a date_month 1

date_second 2

a date_wday 1

date_year 1

a date_zone 1

a index 1

linecount 1

i	Time	Event
>	5/25/26 9:15:52.000 AM	May 25 09:15:52 Multiple failed login attempts from 203.0.113.45 host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs
>	5/25/26 9:10:15.000 AM	May 25 09:10:15 Failed login attempt from 192.168.1.12 host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs

Search | Splunk 10.4.0

localhost:8000/en-US/app/search/search?q=search%20source%3D%20Sample_logs.txt%20host%3D%20Jaya_Kondreddy%20malware&earliest=0&latest=&display.page.search.mode=smart&dispatch...

splunk>Enterprise

New Search

source="Sample_logs.txt" host="Jaya_Kondreddy" malware

Time range: All time

1 event (before 5/25/26 4:12:52.000 PM) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

< Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 1

date_month 1

date_second 1

date_wday 1

date_year 1

date_zone 1

index 1

	i	Time	Event
>		5/25/26 9:12:44.000 AM	May 25 09:12:44 Malware detected on host PC-02 host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs

Search | Splunk 10.4.0

localhost:8000/en-US/app/search/search?q=search%20source%3D%20Sample_logs.txt%20host%3D%20Jaya_Kondreddy%20unauthorized&earliest=0&latest=&display.page.search.mode=smart&dis...

splunk>Enterprise

New Search

source="Sample_logs.txt" host="Jaya_Kondreddy" unauthorized

Time range: All time

1 event (before 5/25/26 4:11:36.000 PM) No Event Sampling

Events (1) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

< Hide Fields

All Fields

SELECTED FIELDS

host 1

source 1

sourcetype 1

INTERESTING FIELDS

date_hour 1

date_mday 1

date_minute 1

date_month 1

date_second 1

date_wday 1

date_year 1

date_zone 1

index 1

linecount 1

	i	Time	Event
>		5/25/26 9:13:27.000 AM	May 25 09:13:27 Unauthorized admin access attempt detected host = Jaya_Kondreddy source = Sample_logs.txt sourcetype = Sample_logs

Search | Splunk 10.4.0

localhost:8000/en-US/app/search/search?q=search%20source%3D%20Sample_logs.txt%20%7C%20stats%20count%20by%20sourcetype&earliest=0&latest=&display.page.search.mode=smart&di...

splunk>Enterprise

New Search

source="Sample_logs.txt" | stats count by sourcetype

Time range: All time

10 events (before 5/25/26 4:19:48.000 PM) No Event Sampling

Events Patterns Statistics (1) Visualization

Chart: Column Chart Format Trellis

sourcetype	count
Sample_logs	10

sourcetype

Sample_logs



RECOMMENDATIONS

- Implement strong authentication methods
- Monitor suspicious login attempts
- Restrict unauthorized access
- Perform regular log monitoring
- Maintain updated security policies



CONCLUSION

This project provided practical exposure to SIEM monitoring, log analysis, and incident response using Splunk Enterprise. The analysis helped in understanding basic SOC operations and cybersecurity monitoring techniques.